

Job Leon Feldbrugge

Univeristy of Edinburgh
Higgs Centre for Theoretical Physics
James Clerk Maxwell Building
Peter Guthrie Tait Road
Edinburgh, EH9 3FD, Scotland

Job.Feldbrugge@ed.ac.uk
jfeldbrugge.github.io

Referees

Prof. Neil Turok, University of Edinburgh, Edinburgh, United Kingdom and Perimeter Institute for Theoretical Physics, Waterloo, Canada (neil.turok@ed.ac.uk)

Prof. Ue-Li Pen, Academia Sinica, Taipei, Taiwan and Canadian Institute for Theoretical Astrophysics, Toronto, Canada (pen@cita.utoronto.ca)

Prof. Rien van de Weygaert, University of Groningen, Groningen, The Netherlands (weygaert@astro.rug.nl)

Academic positions

2021 - present

HIGGS FELLOW at the Higgs Centre for Theoretical Physics at the University of Edinburgh (Scotland)

2019 - 2021

POSTDOC at the Perimeter Institute (Canada) and the Department of Physics, Carnegie Mellon University (United States)

Education

2015 - 2019

PHD Physics, Perimeter Institute, University of Waterloo

Advisor: Neil Turok

Thesis: Path integrals in the sky: Classical and quantum problems with minimal assumptions

Defense date: October 17, 2019

2014 - 2015

MASTER Part III Mathematics (*with distinction*), University of Cambridge

Committee: Paul Shellard and Tommaso Giannantonio

Thesis: Primordial non-Gaussianity and large-scale structure

2012 - 2014

MASTER Physics (*cum laude*), van Swinderen Institute, University of Groningen

2012 - 2014

MASTER Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2012 - 2014

MASTER Mathematics (*cum laude*), Bernoulli Institute, University of Groningen

Committee: Rien van de Weygaert (cosmology and large-scale structure formation)

Diederik Roest (string cosmology)

Aernout van Enter (statistical mechanics)

Thesis: Statistics of caustics in large-scale structure formation

2009 - 2012

BACHELOR Physics (*cum laude*), van Swinderen Institute, University of Groningen

2009 - 2012

BACHELOR Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2009 - 2012

BACHELOR Mathematics (*cum laude*), Bernoulli Institute, University of Groningen

Committee: Rien van de Weygaert (cosmology and large-scale structure formation)

Elisabetta Pallante (quantum field theory)

Gert Vegter (computational geometry)

Thesis: Analysis of Betti numbers and persistence diagrams of 2D Gaussian random fields

July 2016

SUMMER SCHOOL It from qubit summer school, Perimeter Institute, Canada (two weeks)

July 2015

SUMMER SCHOOL Prospects in theoretical physics: new insights into quantum matter, Institute for Advanced Studies, Princeton (one week)

July 2015

SUMMER SCHOOL Princeton summer school on condensed matter physics (one week)

August 2011

SUMMER SCHOOL Dealing with environmental heritage, Bath, United Kingdom (two weeks)

July 2010

SUMMER SCHOOL University of Cambridge international summer school in science (two weeks)

Awards

April 2020	Canadian Association of Physicists' Division of Theoretical Physics (DTP) and Winnipeg Institute for Theoretical Physics (WITP) P.R. Wallace PhD Thesis Prize Best PhD thesis in Theoretical Physics from a Canadian university (2020).
November 2014	De Zeeuw-Van Dishoeck award 2014 Best master thesis in Astronomy from a Dutch university (2014).
July 2014	GUF-100 prize 2014 Best student in the Faculty of Mathematics and Natural Sciences at the University of Groningen (2014).
November 2011	Silver medal in the university physics competition 2011 A university competition in which groups of three physics students all over the world solve a problem and write an article in 48 hours.
September 2010	Young Talent encouragement prize 2010 in Physics Prize awarded by the Koninklijke Hollandse Maatschappij der Wetenschappen (Royal Holland Society of Sciences and Humanities) for the best freshman Physics student at the University of Groningen in 2009-2010.
November 2007	Third place in the CanSat competition The CanSat project is an annual competition organized by the Delft University of Technology. Teams of secondary school students design and build a satellite in a Coca-Cola can.

Scholarships

March 2015	University of Waterloo scholarship, for PhD at the Perimeter Institute
July 2014	Hendrik Muller fund 2014: Scholarship for excelling Dutch students
June 2014	VSB fund 2014: Scholarship for Dutch students studying in abroad

Research Visits

November 2025	Research visit to Chris Howls and Ines Aniceto, Southampton University, Southampton, United Kingdom
October 2025	Research visit to Sandrine Codis, Université Paris-Saclay, Paris, France
September 2025	Research visit to Chris Howls and Ines Aniceto, Southampton University, Southampton, United Kingdom
May 2023	Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
March 2023	Research visit to Ue-Li Pen, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan
March 2023	Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
January 2023	Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2022	Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
January 2020	Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2019	Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
October 2018	Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
May 2018	Research visit to Jean-Luc Lehnert, Albert Einstein Institute, Potsdam, Germany
Feb - Apr 2017	Research visit to DAMTP, University of Cambridge, United Kingdom
March 2017	Research visit to Jean-Luc Lehnert, Albert Einstein Institute, Potsdam, Germany

Publications & talks

JOURNAL ARTICLES

2026	I. Aniceto, J. Feldbrugge , C. J. Howls "Which Saddles Contribute? The South-East Rule for Multidimensional Integrals", (R & R) (2026) arXiv:2606.28271 [math-ph]
2026	A. Read, J. Feldbrugge , C. Boehm, R. van de Weygaert, B. Hertzsch "Caustic Skeleton and the Local Cosmic Web: the Coma Cluster node and the Pisces-Perseus ridge", (R & R) (2026) <i>MNRAS</i> arXiv:2604.22213 [astro-ph.CO]
2026	B. Hertzsch, J. Feldbrugge , R. van de Weygaert "Galaxy Populations in the IllustrisTNG Caustic Skeleton", (R & R) (2026) <i>MNRAS</i> arXiv:2604.18209 [astro-ph.CO]
2026	J. Feldbrugge , S. Crew, P-L Pen, "Resurgence and Hyperasymptotics in Wave Optics Astronomy", (accepted) (2026) <i>JCAP</i> , arXiv:2602.21493 [astro-ph]
2026	A. Weber, J. Feldbrugge , E. Pisanty, "A universal approach to saddle-point methods in attosecond science", (Physical Review A) (2025), arXiv:2510.12545 [quant-ph]

2025 B. Hertsch, **J. Feldbrugge**, Maé Rodriguez and Rien van de Weygaert, “A New Recipe for Caustic Pancakes: On the Reality of Walls in the Cosmic Web”, *JCAP* (2025), [arXiv:2510.02419 \[astro-ph.CO\]](#)

2025 **J. Feldbrugge** and U-L Pen, “The real-time Feynman path integral for step potentials”, *Annals of Physics* (2025), [arXiv:2508.17578 \[quant-ph\]](#)

2025 **J. Feldbrugge** and J. Y. L. Jones, “Efficient evaluation of real-time path integrals”, *Physical Review D* (2025) 111, [arXiv:2501.16323 \[quant-ph\]](#)

2025 B. Bonga, **J. Feldbrugge** and A. Metidieri, “Wave optics for rotating stars”, *Physical Review D* (2025) Volume 111, Issue 6. [arXiv:2410.03828 \[astro-ph.CO\]](#)

2025 **J. Feldbrugge** and R. van de Weygaert, “What makes a cosmic filament? The dynamical origin and identity of filaments I. fundamentals in 2D”, *Monthly Notices of the Royal Astronomical Society* (2025). [arXiv:2405.20475 \[astro-ph.CO\]](#)

2025 **J. Feldbrugge**, “Phase-Space Delaunay Tessellation Field Estimator”, *Monthly Notices of the Royal Astronomical Society* (2025) no.1, 536. [arXiv:2402.16234 \[astro-ph.CO\]](#)

2023 **J. Feldbrugge**, D. L. Jow, U.-L. Pen, “Crossing singularities in the saddle point approximation”, *Physical Review Letters* (2023, R&R). [arXiv:2309.12427 \[quant-ph\]](#)

2023 **J. Feldbrugge**, D. L. Jow, U.-L. Pen, “Complex classical paths in quantum reflections and tunneling”, *Physical Review D* (2025) Volume 111, Issue 8, id.085027, 29 pp. [arXiv:2309.12420 \[quant-ph\]](#)

2023 **J. Feldbrugge**, N.M.D. Niezink, “Orthogonality relations for conical functions of imaginary order”, (2023). [arXiv:2309.05616 \[math\]](#)

2023 **J. Feldbrugge**, “Complex evaluation of angular power spectra: Going beyond the Limber approximation”, *Physical Review D* (2023) nr. 108, 103007. [arXiv:2304.13064 \[astro-ph.CO\]](#)

2023 **J. Feldbrugge**, Y. Yan, and R. van de Weygaert, “Statistics of tidal and deformation eigenvalue fields in the primordial Gaussian matter distribution: the two-dimensional case”, *Monthly Notices of the Royal Astronomical Society* (2023). [arXiv:2301.07200 \[astro-ph.CO\]](#)

2023 **J. Feldbrugge**, and R. van de Weygaert, “Cosmic web & caustic skeleton: non-linear constrained realizations - 2D case studies”, *Journal of Cosmology and Astroparticle Physics* (2013) no.2, 58. [arXiv:2212.07840 \[astro-ph.CO\]](#)

2023 **J. Feldbrugge**, U.-L. Pen, and N. Turok, “Oscillatory path integrals for radio astronomy”, *Annals of Physics* (2023) no.451, 169255. [arXiv:1909.04632 \[astro-ph.HE\]](#)

2023 **J. Feldbrugge**, “Multi-plane lensing in wave optics”, *Monthly Notices of the Royal Astronomical Society* (2023) nr.250, 2995-3006. [arXiv:2010.03089 \[astro-ph.CO\]](#)

2022 **J. Feldbrugge**, and N. Turok, “Existence of real time quantum path integrals”, *Annals of Physics* (2023) [arXiv:2207.12798 \[hep-th\]](#)

2022 D. Jow, U.-L. Pen, and **J. Feldbrugge**, “Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale”, *Monthly Notices of the Royal Astronomical Society* (2022). [arXiv:2204.12004 \[astro-ph.CO\]](#)

2021 G. Wilding, K. Nevenzeel, R. van de Weygaert, G. Vegter, P. Pranav, B.J.T. Jones, K. Efstathiou, and **J. Feldbrugge**, “Persistent homology of the cosmic web. I: Hierarchical topology in Λ CDM cosmologies” *Monthly Notices of the Royal Astronomical Society* (2020) nr.507, 2968-2990. [arXiv:2011.12851 \[astro-ph.CO\]](#)

2020 **J. Feldbrugge**, and N. Turok, “Gravitational lensing of binary systems in wave optics”, *Physical Review Letters* (2020, R&R). [arXiv:2008.01154 \[gr-qc\]](#)

2019 **J. Feldbrugge**, M. van Engelen, R. van de Weygaert, P. Pranav, and G. Vegter, “Stochastic homology of Gaussian vs. non-Gaussian random fields: Graphs towards Betti numbers and persistence diagrams”, *Journal of Cosmology and Astroparticle Physics* (2019) no.9, 52–100. [arXiv:1908.01619 \[astro-ph.CO\]](#)

2019 A. Di Tucci, **J. Feldbrugge**, J.-L. Lehnert, N. Turok, “Quantum incompleteness of inflation”, *Physical Review D*, 100 (2019) no.6, 63517. [arXiv:1906.09007 \[hep-th\]](#)

2019 P. Pranav, R. van de Weygaert, G. Vegter, B.J.T. Jones, R.J. Adler, **J. Feldbrugge**, C. Park, T. Buchert, and M. Kerber, “Topology and geometry of Gaussian random fields I: on Betti numbers, Euler characteristic, and Minkowski functionals” *Monthly Notices of the Royal Astronomical Society*, 485 (2019) no.3, 4167–4208. [arXiv:1812.07310 \[astro-ph.CO\]](#)

2018 **J. Feldbrugge**, J.-L. Lehnert, and N. Turok, “Inconsistencies of the new no-boundary proposal”, *Universe*, 4 (2018), no.10, 100–115. [arXiv:1805.01609 \[hep-th\]](#)

2018 **J. Feldbrugge**, R. van de Weygaert, J. Hidding, and J. Feldbrugge, “Caustic skeleton & cosmic web”, *Journal of Cosmology and Astroparticle Physics* (2018) no.05, 27–81. [arXiv:1703.09598 \[astro-ph.CO\]](#)

2018 **J. Feldbrugge**, J. Lehnert, and N. Turok, “No rescue for the no boundary proposal: Pointers to the future of quantum cosmology”, *Physical Review D*, 97 (2018), no.2, 23509 [arXiv:1708.05104 \[hep-th\]](#)

- 2017 **J. Feldbrugge**, J.-L. Lehnert, and N. Turok, “No smooth beginning for spacetime,” *Physical Review Letters*, 119 (2017), no.17, 171301. [arXiv:1705.00192 \[hep-th\]](#)
- 2017 **J. Feldbrugge**, J.L. Lehnert, and N. Turok, “Lorentzian quantum cosmology,” *Physical Review D*, 95 (2017), no.10, 103508. [arXiv:1703.02076 \[hep-th\]](#)
- 2016 **J. Feldbrugge**, J. Hidding, and R. van de Weygaert “Statistics of caustics in large-scale structure formation,” *The Zeldovich Universe: Genesis and Growth of the Cosmic Web, Proceedings of the International Astronomical Union, IAU Symposium*, 308 (2016), 107–114. [arXiv:1412.5121 \[astro-ph.CO\]](#)
- 2013 R. van de Weygaert, G. Vegter, H. Edelsbrunner, B.J.T. Jones, P. Pranav, C. Park, W. Hellwing, B. Eldering, N. Kruithof, E.G.P. Bos, J. Hidding, **J. Feldbrugge**, E. ten Have, M. van Engelen, M. Caroli, and M. Teillaud, “Alpha, Betti and the megaparsec universe: On the topology of the cosmic web,” *Transactions on Computational Science XIV: Special Issue on Voronoi Diagrams and Delaunay Triangulation. Lecture Notes in Computer Science, Vol. 6970. Springer Berlin Heidelberg* (2013). [arXiv:1306.3640 \[astro-ph.CO\]](#)

THESES

- 2019 **J. Feldbrugge**, “Path integrals in the sky: classical and quantum problems with minimal assumptions,” PhD thesis, Perimeter Institute, University of Waterloo, supervised by N. Turok. [Available online.](#)
- 2015 **J. Feldbrugge**, “Primordial non-Gaussianity and large-scale structure,” Part III Essay, University of Cambridge, supervised by P. Shellard and T. Giannantonio. [Available online.](#)
- 2014 **J. Feldbrugge**, “Statistics of caustics in large-scale structure formation,” Master thesis, University of Groningen, supervised by R. van de Weygaert, D. Roest, A.E. van Enter. [Available online.](#)
- 2012 **J. Feldbrugge** and M. van Engelen, “Analysis of Betti numbers and persistence diagrams of two-dimensional Gaussian random fields,” Bachelor thesis, University of Groningen, supervised by R. van de Weygaert, E. Pallante, G. Vegter. [Available online.](#)

INVITED TALKS

- August 2026 Astronomy Department of The Physics Institute of the Federal University of Rio Grande do Sul
Seminar: Unravelling the cosmic web with caustics
- July 2026 Nancy, France, EXCOSM workshop “Random Structures and Patterns in Spatial Data: Here, There, and Everywhere”
Presentation: The caustic skeleton of the cosmic web
- May 2026 Physics and Astronomy, Nottingham, United Kingdom
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology
- May 2026 The Center of Gravity – Niels Bohr Institute, Copenhagen, Denmark
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology
- April 2026 Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute, Waterloo, Canada
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
- December 2025 King’s College London, London, United Kingdom
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
- November 2025 University of Southampton, Southampton, United Kingdom
Seminar: Unravelling the cosmic web with caustics
- November 2025 University of Vienna, Vienna, Austria
Presentation: Understanding the cosmic web with caustics
- October 2025 Université Paris-Saclay, Paris, France
Presentation: Understanding the cosmic web with caustics
- October 2025 University of Bonn, Bonn, Germany
Presentation: Real-Time Path Integrals, Caustics and Interference
- October 2025 Université Paris-Saclay, Paris, France
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
- September 2025 Cosmology seminar at Perimeter Institute for Theoretical Physics, Waterloo, Canada
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
- August 2025 Seminar at the Canadian Institute for Theoretical Astrophysics, Toronto, Canada
Presentation: Understanding the cosmic web with caustics
- June 2025 The Cosmic Web from Galaxies to Cosmology, Institute for Fundamental Physics of the Universe, Trieste, Italy
Presentation: Caustics and Interference
- March 2025 Seminar at the University of Pennsylvania, Philadelphia, United States,
Presentation: Real-time Path Integrals, Caustics and Interference

October 2024	Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy Presentation: Path integrals in the sky
June 2024	Singular and Oscillatory Integrals, University College London Presentation: Integration in the complex plane with Picard-Lefschetz theory
May 2024	Tuorla-Tartu meeting 2024: Borderless Universe Presentation: What makes a cusp/filament?
December 2023	Gauge-Gravity by the ghats' seminar, Center for High Energy Physics (CHEP), Indian Institute of Science, India. Presentation: Complex classical paths in quantum reflections and tunnelling
December 2023	Large-scale parity violation, Academia Sinica, Institute of Astronomy and Astrophysics, Taipei, Taiwan, Presentation: Dissecting the cosmic web with caustics
October 2023	International Loop Quantum Gravity Seminar Presentation: Complex saddle points in gravitational path integrals
September 2023	Theoretical Physics seminar Newcastle, University of Newcastle Presentation: On the existence of real-time path integrals
September 2023	Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy Presentation: On the existence of real-time path integrals
July 2023	Quantum Gravity 2023, Radboud University, Nijmegen, The Netherlands Presentation: On the existence of real-time path integrals
July 2023	Theoretical physics group, University of New Brunswick, New Brunswick, Canada Presentation: On the existence of real-time path integrals
March 2023	Optimal Transport Theory and Applications to Physics, Ecole Physique, Les Houches, France Presentation: Dissecting the cosmic web with caustics
September 2022	2nd Roman Juskiwicz Symposium, Nicolaus Copernicus Astronomical Center, Warsaw, Poland Presentation: Dissecting the cosmic web with caustics
June 2022	Information Universe 4, University of Groningen, Groningen, The Netherlands Presentation: Dissecting the cosmic web with caustics
May 2022	Cosmology seminar, Oxford University, Oxford, England Presentation: Dissecting the cosmic web with caustics
May 2022	UK Cosmo Meeting 2022, Newcastle University, Newcastle, England Presentation (keynote): Dissecting the cosmic web with caustics
October 2021	Higgs hour, University of Edinburgh, Edinburgh, Scotland Presentation: Interference, caustics and oscillatory integrals
May 2021	Seminar Universidad Nacional Autonoma de Mexico, Mexico city Presentation: Interference phenomena in lensing and quantum physics
May 2021	Sirius A symposium 2021: To infinity and beyond Presentation: The caustic skeleton of the cosmic web
October 2020	Quantum & Gravity Seminar, Radboud Universiteit, Nijmegen, The Netherlands Presentation: Lorentzian quantum cosmology
October 2020	Pusar group meeting, CITA, Toronto, Canada Presentation: Multi-plane lensing and gravitational binary lensing in wave optics
November 2019	Scintillometry 2019, Max Planck Institute for Radio Astronomy, Bonn, Germany Presentation: Oscillatory path integrals for radio astronomy
June 2018	2018 Rotman summer institute in philosophy of cosmology, Godrich, Canada Presentation: Lorentzian quantum cosmology (two lectures)
September 2016	CITA-PI day: Gravitational non-linear instability, CITA, Toronto, Canada Presentation: Shocks in the early universe and gravitational waves

CONTRIBUTED TALKS

June 2025	Cosmology from home 2025 Presentation: What makes a cosmic wall/filament in the caustic skeleton of the cosmic web?
June 2025	Triangular Conference on Cosmological Frontiers in Fundamental Physics 2025, Paris, France Poster: Real-Time Path Integrals
October 2024	Scintillometry 2024, University of Central Florida Presentation: Lensing by rotating stars
July 2024	Relativistic effects and novel observables, University of Geneva Presentation: What makes a filament/wall?
July 2024	Cosmology from home Presentation: What makes a filament/wall?

June 2024	Theoretical Modeling of the Large Scale Structure of the Universe, University of Edinburgh Presentation: What makes a filament/wall?
March 2024	Large Scale Structure group meeting Cambridge, University of Cambridge Presentation: Caustic skeleton: what makes a filament in the cosmic web?
February 2023	The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time, Kavli Institute for Theoretical Physics, UC Santa Barbara, California, United States Presentation: Dissecting the cosmic web with caustics
July 2022	Cosmology from Home Presentation: Dissecting the cosmic web with caustics
March 2022	Cosmic Cartography 2022: Exploring the Cosmic Web and Large-Scale Structure, Kavli IPMU, Kashiwa, Japan Presentation: The caustic web and non-linear constrained Gaussian random fields
October 2021	Tuorla-Tartu meeting 2021, University of Turku, Turku, Finland Presentation: Caustic skeleton of the cosmic web
September 2020	Cosmology from home 2020, virtual conference on all aspects of cosmology. Presentation: The caustic skeleton of the cosmic web
March 2020	Topological statistics group meeting. Department of statistics at Carnegie Mellon University, Pittsburgh, United States Two presentations: Cosmology and topology I, and Cosmology and topology II
February 2020	The centre for the universe Waterloo Centre for Astrophysics day, Waterloo, Canada Presentation: Path integrals for radioastronomy and gravitational lensing II
February 2020	Cosmology group meeting, Perimeter Institute, Waterloo, Canada Presentation: The caustic skeleton of the cosmic web
January 2019	The cosmic web in the local universe, Lorentz center, Leiden, The Netherlands Presentation: The caustics skeleton of the cosmic web
October 2019	The Future of Astronomy, Waterloo Centre for Astrophysics, Waterloo, Canada Poster: Oscillatory path integrals for radio astronomy
September 2019	Theory group meeting, Carnegie Mellon University, Pittsburgh, United States Presentation: Interference and Picard-Lefschetz theory
September 2019	Simplicity III, Perimeter Institute, Waterloo, Canada Presentation: Fun with path integrals II
August 2019	Graduate student meeting, Perimeter Institute, Waterloo, Canada Presentation: Oscillatory integrals in the complex plane
June 2019	Probabilities in cosmology, University of Groningen, The Netherlands Presentation: Lorentzian beginnings of the universe
June 2019	The cosmic web: from galaxies to cosmology, Edinburgh, United Kingdom Presentation: Caustic skeleton of the cosmic web
May 2019	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: lenses and oscillatory integrals
May 2018	Albert Einstein Institute group meeting, Potsdam, Germany Presentation: Classical and weak trajectories
November 2017	Path integral of gravity, Perimeter Institute, Waterloo, Canada Presentation: Quantum incompleteness of inflation II
September 2017	Cosmology group meeting CITA, Canada Presentation: The instability of the no-boundary proposal
July 2017	Cosmic web day, University of Toronto, Toronto, Canada Presentation: The caustic skeleton of the cosmic web
May 2017	PI-day, Perimeter Institute, Waterloo, Canada Presentation: Lorentzian quantum cosmology
May 2017	Theory Canada 12, York University, Toronto, Canada Presentation: Lorentzian quantum cosmology
May 2017	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: Caustics in large-scale structure
May 2017	String cosmology group meeting Van Swinderen Institute, Groningen, Netherlands Presentation: Lorentzian quantum cosmology
April 2017	New Thoughts 3: About the universe and more, Ely, United Kingdom Presentation: Lorentzian quantum cosmology
April 2017	British gravity meeting 2017, University of Oxford, Oxford, United Kingdom Presentation: Lorentzian quantum cosmology

June 2016	Cosmology group meeting CITA, Canada Presentation: Statistics of caustics in large-scale structure
May 2016	Cosmology group meeting Perimeter Institute, Canada Presentation: Statistics of caustics in large-scale structure
March 2016	Statistics of extrema of large-scale structure, Lorentz center, Leiden, The Netherlands Presentation: Statistics of caustics in large-scale structure
June 2014	IAUS 308: The Zel'dovich universe, Tallinn, Estonia Presentation: Statistics of caustics in large-scale structure formation
October 2012	Structure of the cosmic web, Leibniz institute astrophysics, Potsdam, Germany Presentation: Analysis of Betti numbers and persistence diagrams in 2D GRFs

CONFERENCES AND WORKSHOPS (ATTENDED ONLY)

March 2026	Max Planck-IAS-NTU Center Kick-off Conference, Institute for Advanced Science, Princeton, United States
August 2025	Charting the Future Symposium: Big questions in particle physics, strong gravity, and cosmology over the next 25 years, Perimeter Institute, Waterloo, Canada
January 2023	New Directions in Theoretical Physics 4, University of Edinburgh, Edinburgh, Scotland
June 2022	Analogue Models of Gravity and Fluctuation-Induced phenomena, University of Edinburgh, Edinburgh, Scotland
June 2022	Online Workshop "Physics of the Early Universe"
January 2021	Cosmology 2021: the rise of field theory, University of Cambridge, Cambridge, The United Kingdom
October 2020	The information universe: What is the role of information in our Universe? University of Groningen, Groningen, The Netherlands
January 2020	First Dutch Mathematical Relativity Day, Radboud University Nijmegen, Nijmegen, The Netherlands
September 2019	Cosmological frontiers in fundamental physics 2019, Perimeter Institute, Waterloo, Canada
November 2018	Quantum universe, in celebration of Neil Turok's 60th birthday, Centro de Estudio Cientificos (CECs), Valdivia, Chile
August 2018	Cosmology and gravitational physics with lambda, Nordita, Stockholm, Sweden
June 2018	Scanning new horizons: Emergent space-time, black holes and quantum information, Van Swinderen Institute, Groningen, The Netherlands
January 2018	Gravity in the early universe, Princeton University, Princeton, United States
June 2017	Bounce scenarios in cosmology, Perimeter Institute, Waterloo, Canada
January 2017	Fundamentals of the universe, Van Swinderen Institute, Groningen, The Netherlands
January 2017	New directions in theoretical physics II, Higgs center for theoretical physics, Edinburgh, United Kingdom
October 2016	Midwest relativity meeting, Perimeter Institute, Waterloo, Canada
June 2016	Time in cosmology, Perimeter Institute, Canada
June 2016	Concepts and paradoxes in a quantum universe, Perimeter Institute, Canada
June 2016	Cosmological frontiers in fundamental physics 2016, Perimeter Institute, Canada
June 2015	Convergence, Perimeter Institute, Canada
April 2014	Quantum universe 4, University of Groningen, The Netherlands
March 2013	Quantum universe 3, University of Groningen, The Netherlands

Advising

2025 - present	Advisor PhD project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: Evaluating real time path integrals for relativistic systems
2025 - present	Advisor PhD project, Amelie Reader, University of Sydney, Sydney, Australia: Probing cosmological redshift surveys with the caustic skeleton
2023 - present	Advisor PhD project, Benjamin Hertzsch, University of Edinburgh, Edinburgh, United Kingdom: Probing cosmological redshift surveys with the caustic skeleton
2023 - present	Co-advisor PhD project, Anne Weber, University College London, London, United Kingdom: Complex trajectories in laser ionization and recombination experiments
2023 - 2025	Co-advisor PhD project, Johanna Borissova, Perimeter Institute, Waterloo, Canada: Lorentzian worm holes
2023 - 2025	Co-advisor PhD project, Ariadna Metidieri, Radboud University, Nijmegen, The Netherlands: Lensing of rotating stars and photons in wave optics
2022 - present	Co-advisor PhD project, Joshua Jones, Dublin Institute for Advanced Studies, Dublin, Ireland: Real-time path integrals

2024	Advisor master project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: Hamiltonian Monte Carlo sampling of primordial caustic constraints
2023	Advisor master project, Vera Li, University of Edinburgh, Edinburgh, United Kingdom: The caustic skeleton and redshift space distortions
2022 - 2023	Co-advisor bachelor project, Yonatan Sklansky, University of Pennsylvania, United States: Topology of multi-stream regions in N-body simulations
2022 - 2023	Advisor master project, Yihan Yan, University of Waterloo, Canada: Statistics of critical points in eigenvalue fields
2020 - 2022	Co-advisor PhD project, Georg Wilding, University of Groningen, The Netherlands: Topology of the two-dimensional cosmic web
2021 - 2022	Advisor bachelor project, Yihan Yan, Waterloo University, Canada: Homology of two- and three-dimensional Gaussian random fields
2020 - 2021	Co-advisor PhD project, Varun Rustagi, University of Groningen, The Netherlands: The caustic skeleton of the local universe
2018 - 2022	Co-supervisor master project Kevin Bixerman, University of Groningen, The Netherlands: The caustic skeleton of the local universe

Teaching

July 2025	EXCOSM Summer School, Large-scale structure of the Universe: from galaxies to cosmology, Haapsalu, Estonia
June 2023	School on Fundamentals of the Universe, Lorentz Center, Leiden, The Netherlands
June 2018	2018 Rotman summer institute in philosophy of cosmology, Godrich, Canada Two lectures on Lorentzian quantum cosmology
January 2018	PSI winter school, co-supervisor project 'Pair creation in de Sitter spacetime', Huntsville, Canada
2010 - 2015	Physics lecturer, exam training for secondary school students, UOCG Market BV
2013 - 2014	Developer of teaching material for exam training in physics, UOCG Market BV
2008 - 2014	Tutoring in mathematics and physics for secondary school students

Service

2022	Co-organizer of the conference New Directions in Theoretical Physics 4 at the University of Edinburgh
2018 - present	Reviewer for Physical Review Letters (PRL), Physical Review D (PRD), the Journal of Cosmology and Astroparticle Physics (JCAP), the Journal of High Energy Physics (JHEP), Monthly Notices of the Royal Astronomical Society (MNRAS), and Universe.
2013 - 2014	Chairman of the professor Hendrik de Waard foundation
2012 - 2013	Treasurer of the professor Hendrik de Waard foundation
2011 - 2012	Member of the education committee mathematics, chairman of the student council
2010 - 2014	Guide at the Gratama telescope of the Blaauw observatory, Groningen, at stargazing events

Languages

Dutch	Mother tongue	English	Fluent
German	Elementary	French	Elementary
Latin	Elementary		

Programming experience

Mathematica	Fluent	C++	Fluent
Python	Fluent	Matlab	Moderate
Swift	Moderate	Julia	Fluent
Fortran	Moderate	R	Elementary